

Tummy Time for Babies with Large Omphalocele

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Thesis, Purpose, Objective & Approach

How can a hospital-compliant positioning mattress be designed to support safe, long-duration prone positioning for infants with abdominal defects, such as omphalocele, to enhance their muscle development, respiratory function, and overall comfort? The purpose of this project is to address a critical gap in neonatal and pediatric care for infants born with abdominal wall defects, particularly omphalocele. Infants with these conditions often require special care to ensure safe positioning, as the exposed organs complicate typical positioning approaches. Current NICU positioning options do not provide adequate support for prone (tummy-down) positioning, which is essential for motor skill development, shoulder and hip girdle strengthening, and respiratory function [1]. By developing a specialized positioning mattress, this project aims to offer a safer, more comfortable solution for these infants, allowing them to rest in the prone position and potentially reducing time spent in hospital care.

This project contributes to the fields of pediatric care and medical device design by creating a tailored solution for a specific patient population with unique anatomical challenges. While prone positioning is widely recognized for its benefits in typical infant development, there is limited research and few products available that cater to infants with conditions like omphalocele. By developing a mattress that addresses these needs, this project advances knowledge on how specific design considerations can enhance therapeutic positioning for infants with medical conditions that impact traditional care approaches.

The success of this project could have profound implications for hospitals and families.

Improving the comfort and safety of prone positioning for infants with abdominal defects can

support their physical development and increase the likelihood of achieving critical developmental milestones. This device may reduce the need for lengthy hospital stays, thereby easing financial and emotional burdens on families and healthcare systems. Moreover, this project could pave the way for further innovations in neonatal positioning devices for other complex conditions, setting a precedent for design that prioritizes both therapeutic function and hospital safety.

Moving forward, the goal is to refine the prototype into a modular, customizable mattress that can be adapted for each infant's unique anatomy and defect size. The final product will be durable, easy to clean, and compliant with hospital infection control standards. By creating a device that combines ergonomic support with clinical safety, the intended outcome is a widely accessible solution that improves the quality of life for infants with abdominal defects, supports families in their care journey, and strengthens healthcare providers' ability to offer safe, developmentally supportive positioning options in the NICU.

Responsibility

Brian and Isaac are co-leaders. They share responsibilities. Individually, my job will be organizing group meetings, and keeping track of deadlines. We will work together on design thinking and prototyping, research for ideas and prototyping, experimentation, and data analysis of the mattress's performance ensuring it is accompanying the babies' needs.

Timeline

+ Tummy Time Gantt Chart

		November				December				January				February				March				April				May				June			
Task #	Task Title	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4				
1	Introduction																																
2	Research																																
3	Prototyping																																
4	Real Implications																																
5	Tailored Research																																
6	Tailored Prototyping																																
7	Data Analysis																																
8	Finalization																																
9	Publication/Presentation																																

November/December

Get familiar with current and previous research within omphalocele and our advisor's previous work on her project determining inherent constraints in correlation with design goals. Research materials solutions and begin the preliminary design phase with current constraints and potential capabilities of materials in mind. Brainstorm and create multiple first-level designs to select from in the future.

January/February

Begin physical build. Experiment with ideas and finalize a concept. Miniature prototype at first with no function and eventually move to scale. Confirm the idea with expertise to ensure the idea is possible and any possible improvements they may suggest.

March/April

Real implications, meaning proof of concept with a baby with the condition (or at least a baby doll to emulate). Does concept perform well? If not, proceed to tailored research and prototyping to address areas that didn't work.

May/June

We will develop our final product during this time utilizing the last month to perfect our final selected design to meet all necessary constraints by the project proposal.

Itemized Budget

+ Tummy Time Budget

Item	Qty.	Total	Notes
Foam	4	\$813.04	The main idea. This will be for mattress prototyping build
Bean Bag Filler Beads	2	\$100	Playing with another idea of mixing materials to produce support and resistance in different areas of the mattress,
Specialized Mattress (for research purposes)	1	\$130	most likely will buy a seat cushion as it's a smaller version of the mattress. We understand it's properties for stress testing
Specialized Baby Nursing Blankets	1	\$40	This will serve as a bed sheet at first - understand what makes it appropriate for newborns and apply properties to finding which fabric to source
Prototyping Lab space	1	\$225	For prototyping outside of Mentor's space
Fabric + sewing material			To construct bag/bed sheet later
Body Pressure Measurement System (BPMS) - Research	1	NEED TO REQUEST QUOTE	This will be used for prototyping research with a real/doll baby.
		Hopefully remainder covers this^	
Total		\$1085	

References

[1] National Institutes of Health, "Benefits of Tummy Time | Safe to Sleep®,"

<https://safetosleep.nichd.nih.gov/>. <https://safetosleep.nichd.nih.gov/reduce-risk/tummy-time>

[2] G. Huang, L. Ma, and Y. Wu, "Omphaloceles in combination with a self-made hemispherical cushion: a report of 12 cases," *Brazilian Journal of Medical and Biological Research*, vol. 51, p. e7380, Jul. 2018, doi: <https://doi.org/10.1590/1414-431X20187380>.